1

getting data in

It's a Mess

but the mess is the point — today's the easy bit

It's Monday morning. You've started a new job at **StreamWave** as a Data Technician. Your manager Priya hands you **five files** to start with — two CSVs, two Excel workbooks, and one cheeky PDF — and asks for an insight on subscriber churn by the end of next week. (More files will land later in the week for the year-on-year view, but these are the ones to wrangle first.)

Five files. Three formats. Where do I even start?

— you, holding coffee, looking at the desk

The files have **never spoken to each other**. Some IDs have a prefix. Some don't. One file is full of duplicates from a system migration that somebody clearly gave up on. Another is a PDF that, frankly, should have been a spreadsheet.

I'll just merge them. How hard can it be?

spoiler: don't do that.

Good news: today's job is not to clean it, fix it, or analyse it. Today you just have to **get it in** — in a way that won't blow up later, and in a way the assessor can happily watch you do.

BRAIN POWER

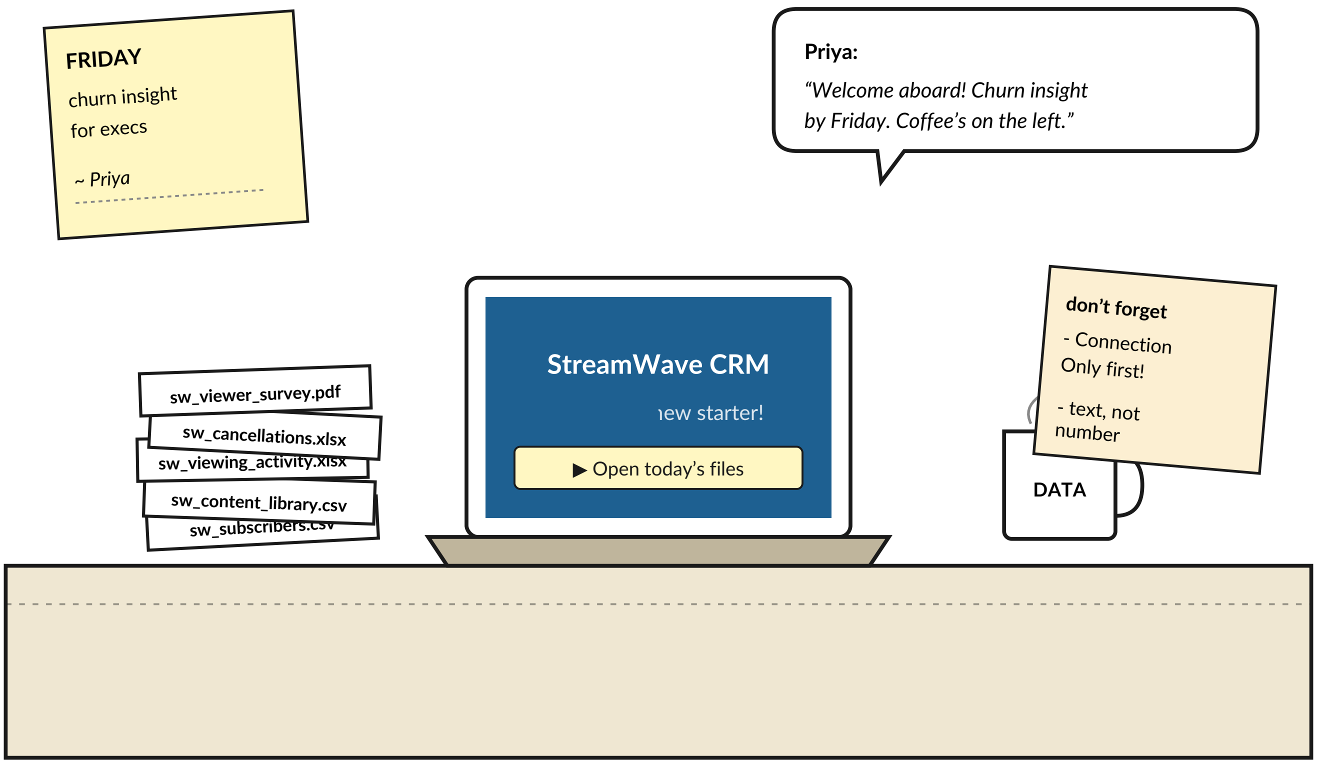
If you just opened every file in Excel and started copy-pasting columns together, what would go wrong? Don't worry about being right — jot down two things you reckon could break before you turn the page.

turn over — your first day starts now →

DAY 1 · 9:03 A.M.

Welcome to StreamWave

Your desk. Your laptop. Your coffee. Your five files. Take a breath — this is what the work actually looks like.



↑ Nothing on the desk has talked to anything else yet. Your job is not to make them talk — it's to **get them into one room** first. That room is Power Query.

Right. Coffee. Then files. We can do this.

Meet the files

Three different file formats — exactly like a real workplace. Power Query handles all of them, but each has its own quirks. Get to know them like co-workers: who's reliable, who needs watching, who's secretly the cause of every problem.

CSV

sw_subscribers.csv — *the friendly one*

One row per subscriber: ID, first & last name, region, plan type, join date, status. The backbone everything else hangs off.

Watch out: subscriber_id looks like a number, isn't one. Keep it as Text or your leading zeros do a runner.

CSV

sw_content_library.csv — *the indecisive one*

Every title on the platform: genre, content type, release year, age rating, average viewer score.

Watch out: genres spelt every which way — Drama, drama, DRAMA. SERIES, Series, series. Years occasionally written as 2022.0.

XLSX

sw_viewing_activity.xlsx — *the chatty one*

3,397 session records: which subscriber watched which title, on what date, for how many minutes, on what device.

Watch out: a few rows have no duration. subscriber_id is a plain integer (1, 2, 3 ...) — the CSV file has it as 5-digit text (00001). Same number, different clothes.

XLSX

sw_cancellations.xlsx — *the one with a past*

55 cancellation records: subscriber ID, cancel date, stated reason, and a migration_flag.

Watch out: rows with migration_flag = Y are duplicates — leftovers from a system migration that ran out of weekend. De-dupe before merging.

PDF

sw_viewer_survey.pdf — *the rebel*

Exit survey responses: overall, content, and value ratings (1–5), a reason code, and a free-text comment.

Watch out: subscriber_id here is `SW00012` — totally different from the CSV (`00012`) and the Excel (`12`). Same person, three different costumes. They will not auto-match.

↑ `subscriber_id` is the key that joins all five files. Three of them store it three different ways. Not on purpose — that's just how data ages in a real company.

The silent killer

Open `sw_subscribers.csv` in Power Query and scroll slowly down the `subscriber_id` column. At first glance everything looks tidy — five-digit numbers, zero-padded, very well-behaved.

But one row in the middle of the file is wearing a slightly different outfit. Miss it now, and when you merge against `sw_viewing_activity` later, that one row will fail to match. The subscriber will disappear from your analysis with **no error and no warning**. Just a missing person.

> *Nothing in data is more dangerous than a failure that doesn't say it failed.*

PREVIEW — SW_SUBSCRIBERS.CSV (ROWS 44–51)

subscriber_id	first_name	last_name	region	plan_type	join_date	status
00044	Becky	Quinn	Yorkshire	Basic	15/08/2021	cancelled
00045	Sinead	Iqbal	London	Basic	13/08/2023	active ·
00046	Beatrice	Lloyd	North West	Premium	20/06/2023	cancelled
SW-00047	**Isla**	**Hussain**	**North West**	**Standard**	**18/05/2022**	**cancelled**
00048	Aidan	Lee	North West	Standard	17/07/2022	active
00049	Brian	Jones	south east	Basic	05/02/2024	active
00050	Tommy	Clark	South West	Standard	17/10/2019	active
00051	Aoife	Lynch	Scotland	Standard	08/09/2022	active

↑ Row 47 — **Isla Hussain** — has the ID `SW-00047`. Prefix and a dash. Everyone else is just digits. Power Query will try to match `SW-00047` against the integer `47`, won't, and quietly drop the row.

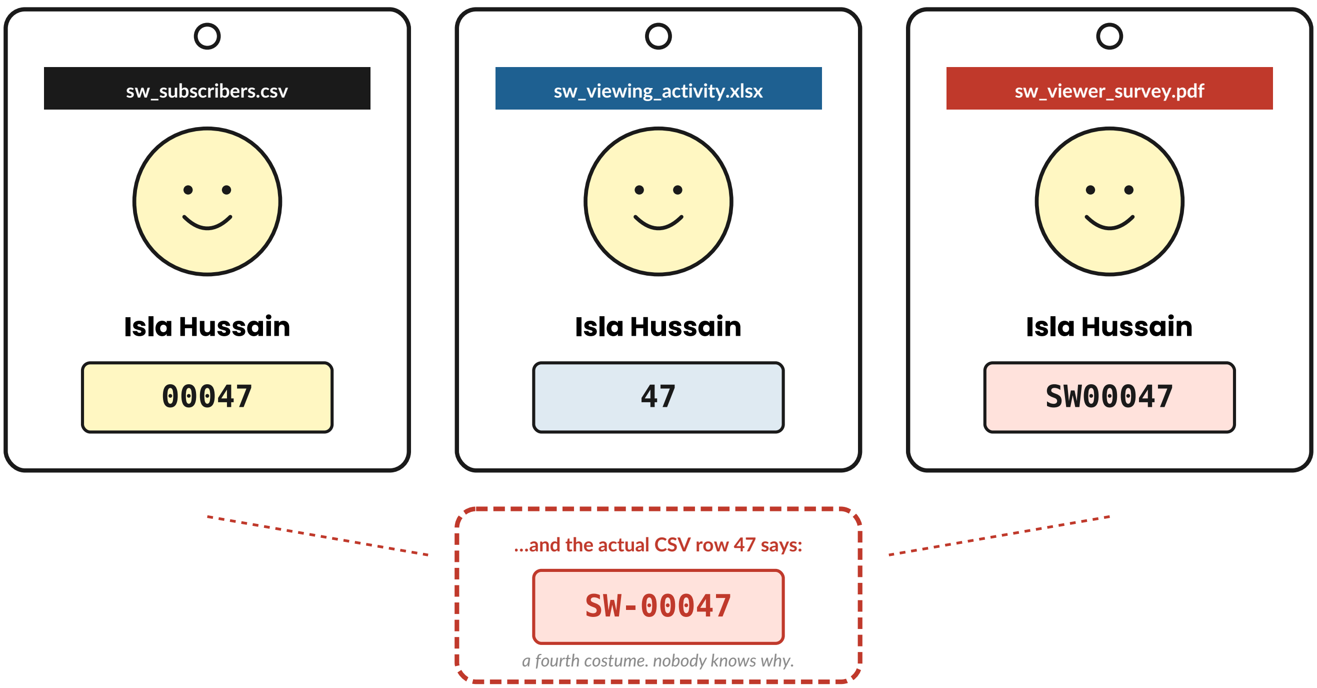
I'm Isla. Miss me and an entire customer goes missing from the churn report.

DON'T FIX IT YET

Today is observation only. Spot it, write it down, say it out loud, move on. The fix happens tomorrow as part of the Cleaning Sequence — where you change every query in the same pass.

Same person. Three costumes.

Subscriber #47 is one human being — Isla Hussain. She shows up in three different files, wearing three completely different name badges. Power Query is many wonderful things, but it is **not** a face-recognition system.



THE RULE WE'LL USE TOMORROW

Pick one canonical format — **five digits, zero-padded, stored as text, no prefix** — and force every source to match it before any merge happens. Once Isla wears the same badge in every file, she stops vanishing.

Halfway House

You've met the files, watched a customer almost vanish from a merge, and seen Isla in her three costumes. Before we look at *how* we actually load these files, let's check what's stuck.

Answer each one true or false from memory. Don't scroll back. Tick the box.

1. Loading every file as Connection Only means the data sits in worksheet tabs until you're ready to analyse it.

☐ **True** ☐ **False**

2. If the join key doesn't match between two queries, Power Query will throw a warning that tells you about it.

☐ **True** ☐ **False**

3. A subscriber_id stored as text in the CSV (`00012`) and as an integer in the Excel (`12`) will merge correctly without any transformation.

☐ **True** ☐ **False**

Answers — check yourself

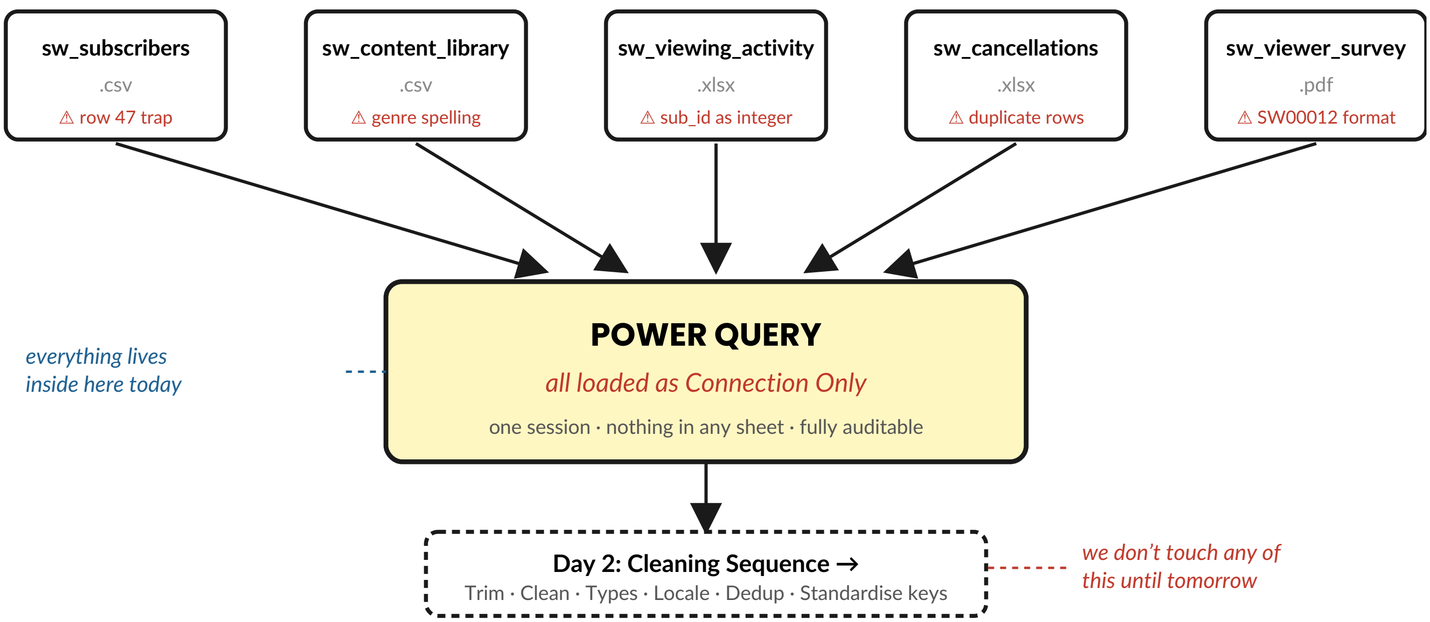
1. FALSE. Connection Only keeps queries inside Power Query. Nothing is written to a worksheet until you load the final, cleaned table.

2. FALSE. Power Query will *not* warn you. It silently drops the rows it can't match. That's exactly why we check keys before merging.

3. FALSE. Different types. You have to standardise both queries first — set the column to Text, strip any prefix, pad to five digits — *in both files*.

The Power Query Pipeline

Every time you bring data into Excel — at work, in the assessment, anywhere — you'll use the same shape. Five files in, one pipeline, every query arriving as **Connection Only**. Learn the picture. The steps come with it.



Why Connection Only first?

If you load a file straight to a sheet, the data lives in *two* places — inside Power Query *and* on a tab. They drift apart the moment you change anything. Connection Only keeps every query inside Power Query, one tidy session, nothing accidentally committed to a cell. It's the difference between a workshop and a junk drawer.

And the killer feature — *Refresh*

Here's the bigger reason we're doing all this in Power Query rather than just opening files in Excel: every step you take today gets **recorded as a repeatable transformation**. Next month, when fresh data arrives, you don't redo any of this — you overwrite the source file, click **Data → Refresh All**, and the entire pipeline reruns automatically. The dashboard you'll build by Friday stays current on one button press, every month, forever. We'll come back to Refresh properly on Day 6 — but it's the reason today's careful work pays off later.

ASSESSOR FAVOURITE

In the post-demo questions, *"Why Connection Only?"* and *"Why Power Query, not just Excel?"* are both first-on-the-list questions. The combined one-sentence answer: *"Connection Only keeps the pipeline in one auditable session, and the whole thing is repeatable on Refresh — so when next month's data lands, the dashboard updates in thirty seconds with no rework."*

The Task

Open a fresh, blank Excel workbook. Your files are in the `StreamWave_EPA_Prep/Data` folder. Don't skip steps and don't multitask — each step builds on the last. Talk to yourself as you go.

1 **Load `sw_subscribers.csv`**

Data tab → **Get Data → From Text/CSV** → pick the file. In the preview window, don't click Transform Data. Click the dropdown arrow next to Load → **Create Connection Only**.

> *"I'm building a pipeline. I'm not loading anything to a sheet yet."*

2 **Repeat for `sw_content_library.csv`**

Same move. Get Data → From Text/CSV → Connection Only. Two queries down, three to go.

3 **Load `sw_viewing_activity.xlsx`**

Get Data → **From Workbook** → pick the file → Connection Only. If Power Query asks which sheet, pick the *viewing_activity* sheet — not any summary tab.

> *"Excel files often have multiple sheets. I always check."*

4 **Load `sw_cancellations.xlsx`**

Same as step 3. Now check the **Queries & Connections** pane on the right. All four should show the little plug icon — not a row count and a sheet icon.

5 **Load `sw_viewer_survey.pdf`**

Get Data → **From PDF** → pick the file → Connection Only. Power Query extracts tables from the PDF. Look at the preview pane on the left. Pick the table that contains the survey columns (*subscriber_id*, *overall*, *content*, *value*, *reason_code*, *free_text_comment*) — not the page header or footer.

> *"If multiple tables show up, I pick the one with the rating columns."*

RED FLAG · MAC USERS

Excel for Mac does not have a *From PDF* connector in Power Query. If you're on a Mac, step 5 will fail — the option simply isn't in the menu. This is a known Microsoft limitation, not something you've done wrong.

Two workarounds. Either (a) run the programme on a Windows machine for the import step only, save the workbook, then continue on your Mac from Day 2 — Power Query refreshes work on Mac once the query exists; or (b) ask a colleague to extract the PDF tables to a CSV first, then load that CSV instead. Note the workaround in your methodology — assessors prefer "I worked around a platform limitation" to silence.

RED FLAG · EXCEL FOR THE WEB

Power Query is read-only in Excel for the Web. You can refresh queries that already exist, but you can't author new ones, edit the M code, or change Applied Steps. For everything in this programme, you need *Excel for Windows desktop*. If your organisation only provides Excel Web, raise it with your line manager before the AM1 date — the assessor will expect you to author Power Query live.

6 Check data types on every column

Right-click `sw_subscribers` → **Edit / Transform Data**. Look at the icon above each column heading: `123` = whole number · `1.2` = decimal · `ABC` = text · the calendar icon = date. Note anything that looks wrong. Don't fix it yet — you're *observing*.

7 Find the `subscriber_id` trap

Scroll slowly down the `subscriber_id` column. Every row follows the same five-digit pattern — except one. Don't filter. Your eye will catch the pattern break before Power Query flags an error.

> *"Found it. Row 47, Isla Hussain — wrong format. If I merged now, she'd silently disappear."*

Quick-fire round

You've just loaded all five files. Before we settle into the pencil exercises, a fast recall drill — read each card, look away, answer out loud in about five seconds. No peeking at earlier pages.

Name the five files.

What format is `subscriber_id` in each of them? (CSV / Excel / PDF)

What's the problem with row 47? With row 45? With row 49?

Why haven't we fixed anything yet?

Stuck on any of them?

That's the whole point — to find out what hasn't stuck yet. Flip back to the relevant page for any you blanked on, then come back. Don't move on until all four feel solid. There's no prize for rushing.

SHARPEN YOUR PENCIL find the trap — under 3 minutes

In the data preview a few pages back you saw eight rows from `sw_subscribers.csv`. Without scrolling back, answer these three questions *from memory*. Then check your answers below the line.

1. Which row contains the `subscriber_id` that won't match?

2. What format is it in (write it out)?

3. What's *exactly* going to happen when Power Query tries to merge it against `sw_viewing_activity`?

Answers — check yourself

1. Row 47 — *Isla Hussain*.

2. `SW-00047` — it has an "SW-" prefix and a dash, where every other row is just five digits like `00046`.

3. Power Query will try to join `SW-00047` against the integer `47` in the viewing file, fail to match, and silently drop the row. Isla disappears from any analysis that depends on the merge — with no error message. The fix (Day 2) is to **standardise** the column: strip any prefix, force to text, and pad to five digits in *every* source.

A SLOWER LOOK — ABOUT 5 MINUTES

Spot the dirt

The `subscriber_id` trap isn't the only thing wrong with that preview. There are at least two other small problems hiding in plain sight. Both will be fixed on Day 2, but today we just want to *notice* them.

SHARPEN YOUR PENCIL two more — spot the dirt

Looking at the same preview, what other small problems can you see in rows 44–51? Write two more before turning the page.

Answers — check yourself

1. Row 45 (*Sinead Iqbal*) — the `status` value has a *trailing space* after "active".

2. Row 49 (*Brian Jones*) — the region is written "south east" in lowercase, where every other row uses Title Case ("South East").

On Day 2 we'll fix both in Power Query: right-click the column → Transform → Trim for the trailing space, and Transform → Format → Capitalize Each Word for the region casing. Two clicks each, no typing.

There are no *Dumb* Questions

Q: *Why bother with Power Query? Couldn't I just open all five files in Excel and use VLOOKUP?*

A: You could — and it would work for one analysis, on one day. The trouble is every time the files refresh you'd have to redo it all manually. Power Query records every step as *code* behind the scenes. Press Refresh and the whole pipeline reruns against the new files. That's the difference between doing the work and doing it once.

Q: *What does "Connection Only" actually mean? Where is the data?*

A: Inside Power Query. Think of it like a recipe written down without cooking it yet. You've told Excel *where* the data is and *what* you'd do to it — but no rows have hit a worksheet. That comes later, when you load the **final, cleaned** table to a sheet for analysis.

Q: *I'm worried I'll mess up the source data. Will Power Query change my original files?*

A: No. Power Query only *reads* source files. Every transformation you apply is stored in the workbook, not the source. You can break a query and the underlying CSV is completely untouched. Breathe.

Q: *The PDF subscriber_id starts with "SW00012", the Excel just has "12", and the CSV has "00012". Which is "right"?*

A: None is more "right" than the others — they're just *different*. What matters is that you pick one canonical format and transform every source to match it before you merge. In the assessment you should **name the mismatch out loud**, decide on a rule ("five digits, zero-padded, text, no prefix"), and apply it consistently. The assessor wants to see the *decision*.

Q: *Why are we observing problems and not fixing them today?*

A: Because fixing without observing is how new bugs are born. Today's job is to load and notice. Day 2 is one careful cleaning pass across every query. Day 3 is the merge. One layer at a time — and you'll never wonder where a bug crept in.

> *Observe today. Clean tomorrow. Merge the day after.*
> *Three quiet steps beat one heroic mess.*

Say it out loud

The assessor isn't only watching *what* you do. They're listening to *why* you did it. Distinction candidates narrate as they work. Practise these three answers *out loud*, right now, as if the assessor were sitting across from you.

Q1 — Why did you load all five files as Connection Only?

Aim for one confident sentence. Yours should include: *auditable*, *single Power Query session*, *nothing committed to a sheet yet*, *repeatable on refresh*.

Try: *Connection Only keeps every query inside Power Query in one auditable session, so the pipeline is repeatable when the source files refresh — and nothing is loaded to a worksheet until I'm ready.*

SHOW THIS OFF

The *Cause* → *Action* → *Consequence* shape is what turns a Pass answer into a Distinction one. Cause: "the IDs don't match across files." Action: "I noted the mismatch before merging." Consequence: "if I'd merged blind, Isla Hussain would have silently disappeared." Three sentences — try saying them out loud now. You'll use this exact pattern every day for the next eight chapters, and on every *why* question the assessor asks.

Q2 — What did you find when you checked `subscriber_id`, and why does it matter?

Describe the format mismatch out loud: row 47, Isla Hussain, is written `SW-00047` — every other row in the CSV is five digits with no prefix. The PDF survey file uses an `SW` prefix without a dash, and the viewing Excel file stores the ID as an integer. If you merge any of these without standardising the key first, Power Query will silently drop rows from the join. Isla, for one, would vanish from your analysis with no error.

← name the column. Name what's wrong. Name the consequence.

Q3 — Why gather data from multiple sources rather than one big file?

Think blind spots. A single file has *systemic* errors — whatever's wrong is wrong everywhere. Multiple sources let you *cross-check*. For StreamWave, the cancellations file tells you who left; the viewing file tells you whether they were watching beforehand. Without both, you can't tell whether price was the real reason for churn — or just the one stated on the way out.

DISTINCTION TELL

Always finish an answer with a *consequence*. Not "I checked the column" but "I checked the column **so that** a key mismatch couldn't silently drop a record in the merge". Cause → action → consequence. That trio is what separates a pass from a distinction.

Bullet Points — the chapter in 30 seconds

- **Five files, three formats.** CSV, XLSX, PDF — Power Query handles all three.
- **Connection Only first.** Everything stays inside Power Query for one auditable session.
- **subscriber_id is the key.** Three files, three different formats — standardise before merging.
- **Observe today, clean tomorrow.** Day 1 finds the problems; Day 2 fixes them systematically.
- **Narrate while you work.** Cause → action → consequence — every time.

TEST DRIVE

Before you close the laptop, run this self-check. Tick each box only when it's true:

- [ ] All five files show as Connection Only in the Queries & Connections pane (plug icon, no row count).
- [ ] Data type checked on every column in every query — anything odd written down.
- [ ] The row 47 anomaly (*Isla Hussain, SW-00047*) identified and explained out loud.
- [ ] The format mismatch between subscribers `(00012)`, viewing `(12)`, and survey `(SW00012)` noted.
- [ ] You can sketch the Power Query Pipeline picture from memory without looking back.

COMING UP — DAY 2

Cleaning with Confidence. You'll fix everything you found today — in one Power Query session, in a fixed order. Text: trim and capitalize via the Transform menu. Numbers: locale and decimal fixes. Dates: regional format correction. De-duplicate the cancellations. Standardise `subscriber_id` across all three formats. By the end of Day 2 you'll have the **Cleaning Sequence** in your head — and your five files will finally speak the same language.

that's Day 1 done — go and have a cup of tea.